

Dedicated KVM Multi-Node Kubernetes Deployment of the Solana Kubernetes Observability Core

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Abstract

This technical report documents the deployment and validation of the Solana Kubernetes Observability Core on a dedicated KVM-based multi-node Kubernetes cluster. The work extends the previous Minikube-based Kubernetes validation by deploying the same observability-oriented Solana workload on a kubeadm cluster composed of one control-plane virtual machine and two worker virtual machines. The validated deployment includes a Solana validator StatefulSet, validator ledger PersistentVolumeClaim, wallet-init Job, latency monitor Deployment, RPC Service, Yellowstone/Geyser gRPC Service, metrics Service, and wallet transfer validation Job. The stage remains focused on infrastructure and observability.

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1 Introduction

The Solana Containerised Testbed is a reproducible local Solana environment for infrastructure, observability, dataset generation, and latency/performance research on legacy x86_64 hardware.

This report documents the deployment of the Solana Kubernetes Observability Core on a dedicated KVM-based multi-node Kubernetes cluster.

2 Project Evolution

The project evolved through several stages:

1. Compose-based local Solana testbed.
2. No-AVX2 compatibility workflow.
3. Yellowstone/Geyser observability extension.
4. Minikube-based Kubernetes validation.
5. Dedicated KVM multi-node Kubernetes deployment.

3 Host and Virtualisation Environment

The deployment target is a CentOS Stream 9 host with KVM/libvirt support. VM disks are placed under the host home filesystem rather than the smaller root filesystem.

4 Kubernetes Cluster Topology

The validated cluster uses one control-plane VM and two worker VMs.

Table 1: Dedicated KVM Kubernetes cluster topology.

Node	Role	Label	Main purpose
k8s-cp-01	control-plane	testbed-role= control-plane	Kubernetes API and control-plane services.
k8s-worker-01	worker	testbed-role=validator	Solana validator workload and validator ledger PVC.
k8s-worker-02	worker	testbed-role= observability	Monitor workload and auxiliary validation jobs.

5 Kubernetes Bootstrap and Ansible Automation

The infrastructure automation is stored under:

```
infra/ansible/kvm-kubeadm/
```

The automation covers node preparation, containerd installation, Kubernetes tool installation, control-plane initialisation, worker join, CNI installation, node labelling, local-path storage installation, validation, and reset support.

6 Solana Kubernetes Observability Core

The portable Kubernetes manifests are stored under:

```
k8s/base/
```

The deployed core includes the validator StatefulSet, validator ledger PVC, wallet-init Job, monitor Deployment, RPC Service, Yellowstone/Geyser gRPC Service, and metrics Service.

7 Kustomize Overlay for KVM Multi-Node Deployment

The dedicated KVM multi-node overlay is stored under:

```
k8s/overlays/kvm-multinode/
```

The overlay adds validator placement on `testbed-role=validator`, monitor placement on `testbed-role=observability`, and explicit `local-path` storage for the validator ledger PVC.

8 Validation Evidence

The validation evidence is stored under:

```
results/kvm-kubeadm-cluster/  
results/kvm-multinode-observability-core/
```

The RPC health check returned:

```
{"jsonrpc": "2.0", "result": "ok", "id": 1}
```

After wallet transfer validation, the metrics endpoint reported:

```
solana_transaction_latency_seconds_count 1
```

9 Scope and Limitations

This report documents infrastructure and observability validation only. Load generation, dashboarding, Prometheus/Grafana integration, MPC, reinforcement learning, MARL, and Agave are out of scope.

10 Conclusion

The Solana Kubernetes Observability Core has been successfully deployed and validated on a dedicated KVM-based multi-node Kubernetes cluster.

References